THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Quadrant I	Check the signs of the abscissa and ordinate.
2	**(b)** 6	The abscissa is the x-coordinate.
3	**(b)** -6	The ordinate is the y-coordinate.
4	**(d)** 5	$\sqrt{3^2+4^2}=5$
5	**(c)** $(4,6)$	Average the abscissae and the ordinates.
6	**(b)** ordinate 0	Recall the properties of the axes.
7	**(a)** $(3,-4)$	Only the sign of the ordinate changes.
8	**(c)** 7	Same ordinate: distance $=\lvert x_2-x_1\rvert$.
9	**(c)** Quadrant II	Check the signs of the abscissa and ordinate.
10	**(c)** -7	The abscissa is the x-coordinate.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Abscissa $=4$, ordinate $=3$ (1); Point $=(4,3)$ in Quadrant I (1).	1+1
12	$d=\sqrt{6^2+8^2}=\sqrt{100}$ (1); $d=10$ units (1).	1+1
13	$\left(\frac{2+6}{2},\frac{10+4}{2}\right)$ (1); Midpoint $=(4,7)$ (1).	1+1
14	Abscissa $=-5$, ordinate $=2$ (1); Point $=(-5,2)$ in Quadrant II (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB=\sqrt{(4-(1))^2+(6-(2))^2}=\sqrt{(3)^2+(4)^2}$ (1); $AB=\sqrt{9+16}=\sqrt{25}$ (1); $AB=5$ units (1).	1+1+1
16	Signs $(+,+)$: Quadrant I (1); Reflection in x-axis: $(5,-8)$ (1); Reflection in y-axis: $(-5,8)$ (1).	1+1+1
17	Midpoint $=(1,2)$ (1); $d=\sqrt{(1)^2+(2)^2}$ (1); $d=\sqrt{5}$ units (1).	1+1+1
	OR: $AB=\sqrt{(3-(-2))^2+(13-(1))^2}=\sqrt{(5)^2+(12)^2}$ (1); $AB=\sqrt{25+144}=\sqrt{169}$ (1); $AB=13$ units (1).	1+1+1
18	Signs $(-,+)$: Quadrant II (1); Reflection in x-axis: $(-4,-6)$ (1); Reflection in y-axis: $(4,6)$ (1).	1+1+1
	OR: Midpoint $=(4,2)$ (1); $d=\sqrt{(4)^2+(2)^2}$ (1); $d=2\sqrt{5}$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB=\sqrt{15^2+0^2}=15$ (1); $BC=\sqrt{0^2+20^2}=20$ (1); $AC=\sqrt{225+400}=25$ (1); $AB^2+BC^2=225+400=625$ (1); $AC^2=625$, so $AB^2+BC^2=AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant I (1); Distance from x-axis $=\lvert 4\rvert=4$; from y-axis $=\lvert 3\rvert=3$ (1); $OP=\sqrt{9+16}=5$ (1); Reflection in the origin: $(-3,-4)$ (1); The origin is the midpoint of P and its image: $\left(\frac{3+(-3)}{2},\frac{4+(-4)}{2}\right)=(0,0)$ (1).	1+1+1+1+1
20	$AB=\sqrt{10^2+0^2}=10$ (1); $BC=\sqrt{0^2+24^2}=24$ (1); $AC=\sqrt{100+576}=26$ (1); $AB^2+BC^2=100+576=676$ (1); $AC^2=676$, so $AB^2+BC^2=AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant II (1); Distance from x-axis $=\lvert 12\rvert=12$; from y-axis $=\lvert -5\rvert=5$ (1); $OP=\sqrt{25+144}=13$ (1); Reflection in the origin: $(5,-12)$ (1); The origin is the midpoint of P and its image: $\left(\frac{-5+(5)}{2},\frac{12+(-12)}{2}\right)=(0,0)$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the point as (y,x): the abscissa x always comes first, so $(2,3)\neq(3,2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3,4)$ in Quadrant IV; Quadrant II is $(-,+)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0,-6)$ or $(5,0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	The ordinate is the y-coordinate.
2	**(a)** 13	$\sqrt{5^2+12^2}=13$
3	**(c)** $(6,6)$	Average the abscissae and the ordinates.
4	**(b)** abscissa 0	Recall the properties of the axes.
5	**(a)** $(-2,-5)$	Only the sign of the ordinate changes.
6	**(d)** 7	Same ordinate: distance $=\lvert x_2-x_1\rvert$.
7	**(b)** Quadrant III	Check the signs of the abscissa and ordinate.
8	**(c)** 4	The abscissa is the x-coordinate.
9	**(a)** 8	The ordinate is the y-coordinate.
10	**(d)** 17	$\sqrt{8^2+15^2}=17$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d=\sqrt{9^2+12^2}=\sqrt{225}$ (1); $d=15$ units (1).	1+1
12	$\left(\frac{8+4}{2},\frac{2+14}{2}\right)$ (1); Midpoint $=(6,8)$ (1).	1+1
13	Abscissa $=-3$, ordinate $=-6$ (1); Point $=(-3,-6)$ in Quadrant III (1).	1+1
14	$d=\sqrt{20^2+21^2}=\sqrt{841}$ (1); $d=29$ units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB=\sqrt{(8-(0))^2+(12-(-3))^2}=\sqrt{(8)^2+(15)^2}$ (1); $AB=\sqrt{64+225}=\sqrt{289}$ (1); $AB=17$ units (1).	1+1+1
16	Signs $(-,-)$: Quadrant III (1); Reflection in x-axis: $(-9,2)$ (1); Reflection in y-axis: $(9,-2)$ (1).	1+1+1
17	Midpoint $=(4,4)$ (1); $d=\sqrt{(4)^2+(4)^2}$ (1); $d=4\sqrt{2}$ units (1).	1+1+1
	OR: $AB=\sqrt{(11-(4))^2+(23-(-1))^2}=\sqrt{(7)^2+(24)^2}$ (1); $AB=\sqrt{49+576}=\sqrt{625}$ (1); $AB=25$ units (1).	1+1+1
18	Signs $(+,-)$: Quadrant IV (1); Reflection in x-axis: $(3,7)$ (1); Reflection in y-axis: $(-3,-7)$ (1).	1+1+1
	OR: Midpoint $=(1,6)$ (1); $d=\sqrt{(1)^2+(6)^2}$ (1); $d=\sqrt{37}$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB=\sqrt{9^2+0^2}=9$ (1); $BC=\sqrt{0^2+40^2}=40$ (1); $AC=\sqrt{81+1600}=41$ (1); $AB^2+BC^2=81+1600=1681$ (1); $AC^2=1681$, so $AB^2+BC^2=AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant III (1); Distance from x-axis $=\lvert-15\rvert=15$; from y-axis $=\lvert-8\rvert=8$ (1); $OP=\sqrt{64+225}=17$ (1); Reflection in the origin: $(8,15)$ (1); The origin is the midpoint of P and its image: $\left(\frac{-8+(8)}{2},\frac{-15+(15)}{2}\right)=(0,0)$ (1).	1+1+1+1+1
20	$AB=\sqrt{12^2+0^2}=12$ (1); $BC=\sqrt{0^2+35^2}=35$ (1); $AC=\sqrt{144+1225}=37$ (1); $AB^2+BC^2=144+1225=1369$ (1); $AC^2=1369$, so $AB^2+BC^2=AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant IV (1); Distance from x-axis $=\lvert-24\rvert=24$; from y-axis $=\lvert7\rvert=7$ (1); $OP=\sqrt{49+576}=25$ (1); Reflection in the origin: $(-7,24)$ (1); The origin is the midpoint of P and its image: $\left(\frac{7+(-7)}{2},\frac{-24+(24)}{2}\right)=(0,0)$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the point as (y,x): the abscissa x always comes first, so $(2,3)\neq(3,2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3,4)$ in Quadrant IV; Quadrant II is $(-,+)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0,-6)$ or $(5,0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.

- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $(6, 4)$	Average the abscissae and the ordinates.
2	**(b)** $(0, 0)$	Recall the properties of the axes.
3	**(b)** $(6, 1)$	Only the sign of the ordinate changes.
4	**(b)** 5	Same ordinate: distance $= \lvert x_2 - x_1 \rvert$.
5	**(a)** Quadrant IV	Check the signs of the abscissa and ordinate.
6	**(c)** -8	The abscissa is the x-coordinate.
7	**(c)** -4	The ordinate is the y-coordinate.
8	**(c)** 25	$\sqrt{7^2 + 24^2} = 25$
9	**(d)** $(4, 2)$	Average the abscissae and the ordinates.
10	**(a)** 4	Recall the properties of the axes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\left(\frac{6+10}{2}, \frac{6+2}{2}\right)$ (1); Midpoint $= (8, 4)$ (1).	1+1
12	Abscissa $= 7$, ordinate $= -4$ (1); Point $= (7, -4)$ in Quadrant IV (1).	1+1
13	$d = \sqrt{12^2 + 16^2} = \sqrt{400}$ (1); $d = 20$ units (1).	1+1
14	$\left(\frac{0+8}{2}, \frac{12+4}{2}\right)$ (1); Midpoint $= (4, 8)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = \sqrt{(3-(-3))^2 + (6-(-2))^2} = \sqrt{(6)^2 + (8)^2}$ (1); $AB = \sqrt{36 + 64} = \sqrt{100}$ (1); $AB = 10$ units (1).	1+1+1
16	Signs $(-, +)$: Quadrant II (1); Reflection in x-axis: $(-1, -10)$ (1); Reflection in y-axis: $(1, 10)$ (1).	1+1+1
17	Midpoint $= (6, 4)$ (1); $d = \sqrt{(6)^2 + (4)^2}$ (1); $d = 2\sqrt{13}$ units (1).	1+1+1
	OR: $AB = \sqrt{(11-(2))^2 + (17-(5))^2} = \sqrt{(9)^2 + (12)^2}$ (1); $AB = \sqrt{81 + 144} = \sqrt{225}$ (1); $AB = 15$ units (1).	1+1+1
18	Signs $(+, -)$: Quadrant IV (1); Reflection in x-axis: $(8, 5)$ (1); Reflection in y-axis: $(-8, -5)$ (1).	1+1+1
	OR: Midpoint $= (7, 4)$ (1); $d = \sqrt{(7)^2 + (4)^2}$ (1); $d = \sqrt{65}$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB = \sqrt{18^2 + 0^2} = 18$ (1); $BC = \sqrt{0^2 + 24^2} = 24$ (1); $AC = \sqrt{324 + 576} = 30$ (1); $AB^2 + BC^2 = 324 + 576 = 900$ (1); $AC^2 = 900$, so $AB^2 + BC^2 = AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant II (1); Distance from x-axis $= \lvert 8 \rvert = 8$; from y-axis $= \lvert -6 \rvert = 6$ (1); $OP = \sqrt{36 + 64} = 10$ (1); Reflection in the origin: $(6, -8)$ (1); The origin is the midpoint of P and its image: $\left(\frac{-6+(6)}{2}, \frac{8+(-8)}{2}\right) = (0, 0)$ (1).	1+1+1+1+1
20	$AB = \sqrt{16^2 + 0^2} = 16$ (1); $BC = \sqrt{0^2 + 30^2} = 30$ (1); $AC = \sqrt{256 + 900} = 34$ (1); $AB^2 + BC^2 = 256 + 900 = 1156$ (1); $AC^2 = 1156$, so $AB^2 + BC^2 = AC^2$: right angle at B (1).	1+1+1+1+1
	OR: Quadrant: Quadrant I (1); Distance from x-axis $= \lvert 12 \rvert = 12$; from y-axis $= \lvert 9 \rvert = 9$ (1); $OP = \sqrt{81 + 144} = 15$ (1); Reflection in the origin: $(-9, -12)$ (1); The origin is the midpoint of P and its image: $\left(\frac{9+(-9)}{2}, \frac{12+(-12)}{2}\right) = (0, 0)$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
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